Contents

Chapter 3

Functions

3.1 Relations

Definition 3.1.1. A relation R from a set X to a set Y is a set of ordered pairs (x, y) such that for each $x \in X$, there corresponds atleast one $y \in Y$.

We write xRy to mean that x is related y. We shall call X the *input* set and Y the *output* set.

Generally speaking, a relation is simply a rule which connects two elements from different sets.

3.1.2 Classification of Relations

Let R be a relation from X to Y.

Relations for which each element of the set X is mapped to a unique element of the set Y are said to be *one-to-one.*

Example 3.1.3. Let $X = \{3, 5, 7\}$ and $Y = \{6, 10, 14\}$, and let the relation R from X to Y be defined by : " is a factor of". Display this on an arrow diagram. Clearly, this is a one-to-one.

A relation can map more than one element of the set X to the same element of the set Y. Such a type of relation is said to be *many-to-one.*

Example 3.1.4. Let $X = \{4, 5, 6\}$ and $Y = \{2\}$, and let the relation R from X to Y be defined by : " is greater than". Display this on an arrow diagram. Clearly, this is a *many-to-one.*

A relation can map one element of the set X to more than one element of the set Y. Such a type of relation is said to be *one-to-many.*

Example 3.1.5. Let R be a relation from X to Y defined by $'<'$ given that $X = \{1, 2, 3\}$ and $Y = \{4, 3, 5\}$. Display this on an arrow diagram. Clearly, this is a *one-to-many.*

The inverse of the relation $'<'$, defined in the previous example is called $'>'$: and is given by

$$R^{-1} = \text{ inverse of } R = \{(3,1),(3,2),(4,1),(4,2),(4,3),(5,1),(5,2),(5,3)\}.$$

3.1.6 Domain, Range and Co-domain

Let R be a relation from X to Y.

1. The set of all values for which the relation is defined (set of input values) is called the domain (D).
2. The range (R) is the set of all values that it can produce. It is the set of output values.
3. The co-domain of a relation is a set of values that includes the range as described above, but may also include additional values beyond those in the range.

Example 3.1.7. Let R be a relation from X to Y defined by $'<'$ given that $X = \{1, 2, 3, 4\}$ and $Y = \{1, 3, 5\}$. Display this on an arrow diagram and hence find the domain, range and the co-domain.

3.2 Functions

Definition 3.2.1. A function from a set X to a set Y is a set of ordered pairs (x, y) such that for each $x \in X$, there corresponds a unique $y \in Y$.

Clearly, a function is a relation but the converse is not true. The domain, range and the co-domain for a function are defined as before.

Throughout this course, all functions will be defined on the subset of $\mathbb{R}$. That is; the domain will be the set of real numbers.

Since a function assigns every element $x \in X$ to exactly one element in set Y, we will denote the output elements (images) by $y = f(x)$, where x is the input and is called the *independent variable* and y is called the *dependent variable*. $f(x)$ is read as "f of x" or the value of f at x.

Example 3.2.2. The function

$$f(x) = \frac{x}{2} + 7$$

is the rule that takes a number, divides it by 2, and then adds 7 to the quotient. For example, if $x = 4$, $f(4) = 9$.

3.2.3 Classification of functions

Let f be a function from X to Y.

Functions for which each element of the set X is mapped to a unique element of the set Y are said to be *one-to-one.*

Example 3.2.4. Let $X = \{3, 5, 7\}$ and $Y = \{6, 10, 14\}$, and let the relation R from X to Y be defined by : " is a factor of". Display this on an arrow diagram. Since every input element has a unique image, this relation is a function. Clearly this is *one-to-one.*

A function can map more than one element of the set X to the same element of the set Y. Such a type of function is said to be *many-to-one.*

Example 3.2.5. Let $X = \{4, 5, 7\}$ and $Y = \{3\}$, and let the relation R from X to Y be defined by : " is greater than".Display this on an arrow diagram. Clearly, this is a *many-to-one.*

Definition 3.2.6. A function f is *one-to-one* if and only if $f(a) = f(b)$ for any a and b in the domain means $a = b$.

Example 3.2.7. Let $f(x) = 3x + 1$. Show that f is *one-to-one.*

Example 3.2.8. Which of the following are functions:

1. $y = f(x) = -2x + 7$
2. $y^2 = x$
3. $y = 4$
4. $y = x^2$.

Example 3.2.9. 1. Given that $f(x) = x^2 + 5x - 6$, find $f(3)$ and $f(-3)$.

2. Given that $f(x) = \frac{4x^2-9x+17}{x+7}$, find $f(5)$ and $f(-4)$.

Example 3.2.10. For each of the following functions, find the domain.

1. $y = 4x^2 + 7x - 19$
2. $y = \sqrt{t-5}$
3. $y = \frac{7}{x(x-4)}$
4. $y = \frac{6x}{(x-5)(x-9)}$

Example 3.2.11. Find the range of the following functions given that $-2 \leq x \leq 2$.

1. $y = 2x$
2. $y = x^2$
3. $y = \frac{1}{1-x}$

Listed below are examples of different functions.

1. Linear: $f(x) = ax + c,\ a, c \in \mathbb{R}$.
2. Quadratic: $f(x) = ax^2 + bx + c,\ a, b, c \in \mathbb{R}, a \neq 0$.
3. Polynomial: $f(x) = a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1 x + a_0$.
4. Rational: $f(x) = \frac{x^2-9}{x+4},\ x \neq -4$.
5. Exponential: $f(x) = a^x,\ \ a \neq 0$.
6. Logarithmic function $f(x) = \log_a x$

3.2.12 Inverse of a function

An inverse function is a function that " reverses" another function. This means that if the function f applied to an input x gives a result of y, then applying its inverse function g to y gives the result x, and vice versa. That is $f(x) = y$ and iff $g(y) = x$.

Example 3.2.13. Find the inverse of each of the following functions, and in each case state the range of f and the domain of f^{-1}.

1. $f(x) = 3x + 1$
2. $f(x) = \frac{x-1}{x+2}, \ x \neq -2.$
3. $f(x) = \frac{1}{x-1}, \ x \neq 1.$

3.2.14 Composite functions

Definition 3.2.15. Given two functions f and g, the composite function, denoted by fog (read f composed with g" or "f of g"), is defined by

$$(f \ og) = f(g(x)).$$

In the definition above, we refer to g as the inside function and f as the outside function.

When determining the domain for the composite function, the domain for the inside function and the domain for the resultant composite function must be accounted for. That is; the domain for the composite function is the intersection of the domain of the inside function and that of the resultant composite function.

Example 3.2.16. Let f and g be two functions defined by $f(x) = 2x - 1$ and $g(x) = 3x$. Find

1. $(gof)(x)$
2. $(fog)(x)$

3. $(fof)(x) = f^2(x)$

4. $(gog)(x) = g^2(x)$

Example 3.2.17. Find $(g\ of)(x)$ if $f(x) = \frac{5}{x+4}$ and $g(x) = \frac{3x}{2x-1}$. State the domain of gof.

3.2.18 Odd and even functions

Definition 3.2.19. An even function is one which is unchanged when the sign of its argument changes, $f(-x) = f(x)$.

Examples of such function include $f(x) = x^2 + 2$, $f(x) = 3x^4 + x^2$. Thus polynomials with even powers are even functions.

Definition 3.2.20. An odd function is a function such that $f(-x) = -f(x)$.

Examples of odd function include $f(x) = 3x$, $f(x) = 2x^3 - x$, that is polynomials with odd powers of x.

Example 3.2.21. State whether each of the following functions is even, odd or neither.

1. $f(x) = \frac{x^2+4}{x^3-x}$

2. $f(x) = x^4 - 3x^2 + 7$